

Binary Reverse Engineering Engineer Assessment

Stack Buffer Overflow Analysis Question

Total Points: 20 (No Partial Credit - All or Nothing)

Question: Binary Analysis and Buffer Overflow Exploitation

Your Task

You are provided with a compiled binary that prompts for user input. Reverse engineer it to discover how to trigger the hidden message: **"Be the change!"**

Required Analysis Steps

1. **Identify the vulnerability** using disassembler/decompiler
2. **Use debugger** to find exact buffer size and variable locations
3. **Calculate overflow payload** to overwrite the target variable
4. **Execute successful exploit** showing the target message

Required Deliverables

1. Critical Analysis Screenshots

- Disassembler showing the vulnerable input function and program logic
- Debugger showing stack layout with variable positions
- Memory view showing overflow in progress

2. Successful Exploitation

- Final payload calculation (exact byte count needed)
- Screenshot showing "Be the change!" message displayed

Submission Requirements

- **Files:** Analysis screenshots and payload documentation
- **Subject:** "Buffer Overflow Analysis - [Your Name] - [Timestamp]"

MANDATORY SCREENSHOT EVIDENCE

 **REQUIRED SCREENSHOTS:**

- Disassembler view showing vulnerable function and target message
- Debugger showing stack variable locations and addresses
- Successful execution displaying "Be the change!" message
- Payload calculation showing exact buffer overflow size needed

Failure to provide working exploit showing the target message will result in automatic failure.

Scoring: 20 Points (All or Nothing) **Prerequisites:** Reverse engineering skills, debugger usage, buffer overflow concepts